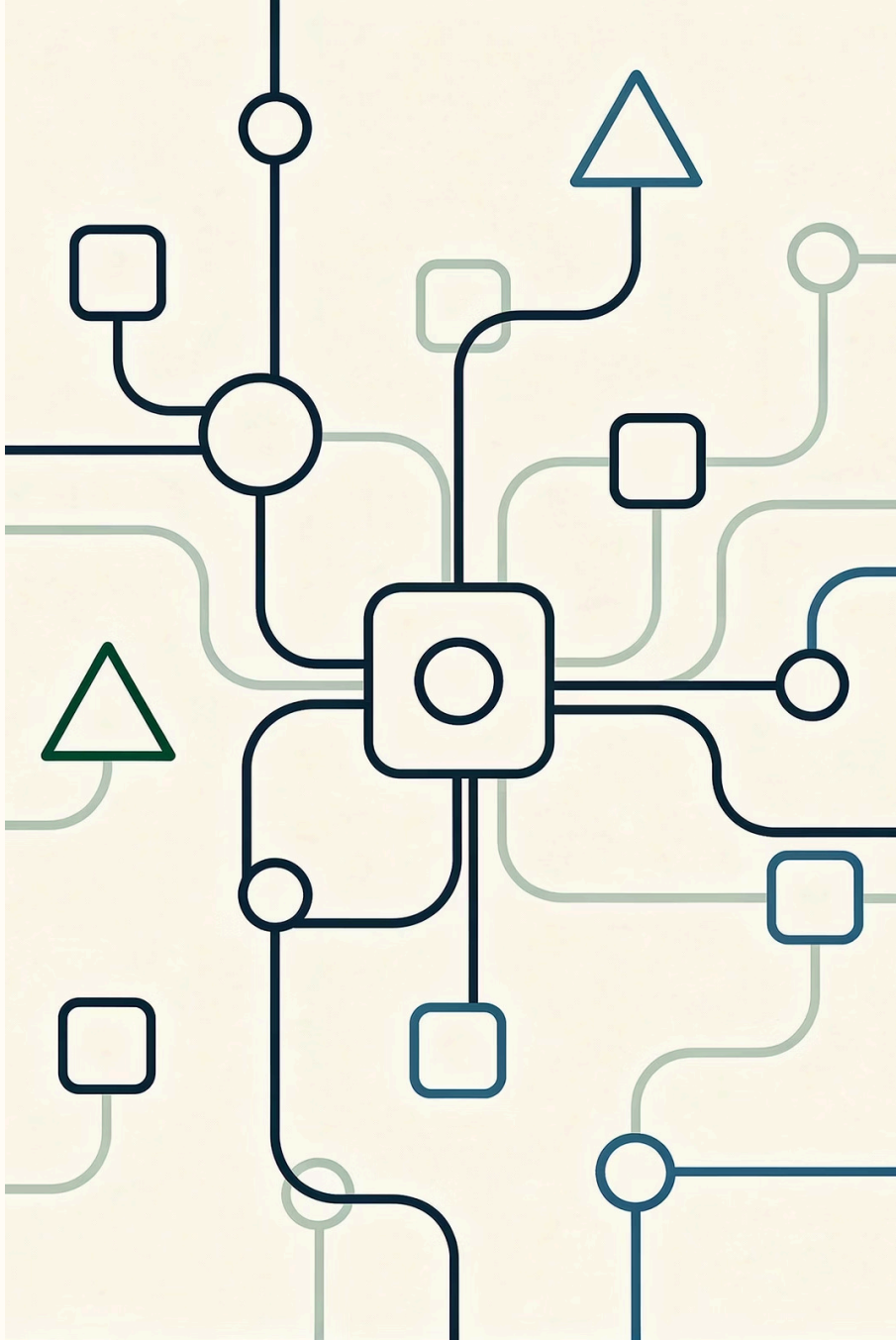
Why Interoperability Roadmaps Fail — and How to Fix Them

Common pitfalls in interoperability strategy and a framework for building roadmaps that actually deliver results.



⚠ THE PROBLEM

The High Cost of Failure: Why Interoperability Roadmaps Matter

Despite billions invested and decades of effort, fewer than half of hospitals routinely share usable data across systems in 2025. The consequences are severe: fragmented patient care, duplicated tests, wasted resources, and missed opportunities for better clinical outcomes.

This page reveals the most common pitfalls that derail interoperability roadmaps — and provides a proven framework for building strategies that deliver real, measurable results.

70%

Engage — But Rarely Routinely

of U.S. hospitals participate in all four interoperability domains, but often only 'sometimes' (ONC/ASTP, 2024)

20–40%

Healthcare Spend Wasted

of the $4.3T annual U.S. healthcare spend is classified as waste tied to fragmentation (Industry estimates, 2025)

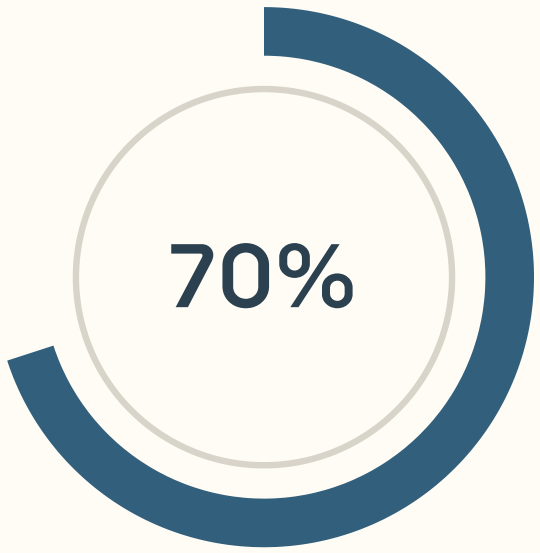
10+

Years of Effort

Years of federal policy effort — yet only 16% of hospitals share care records with most post-acute providers (HealthIT.gov, 2024)

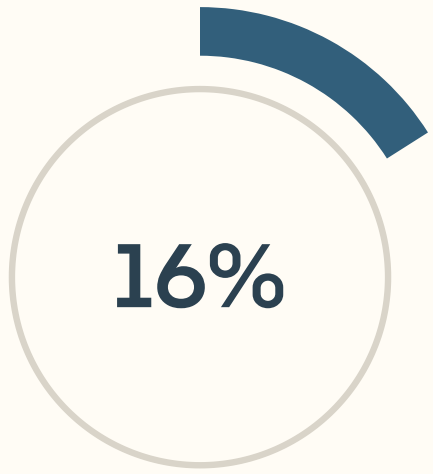
KEY DATA

The Evidence Is Clear: Interoperability by the Numbers



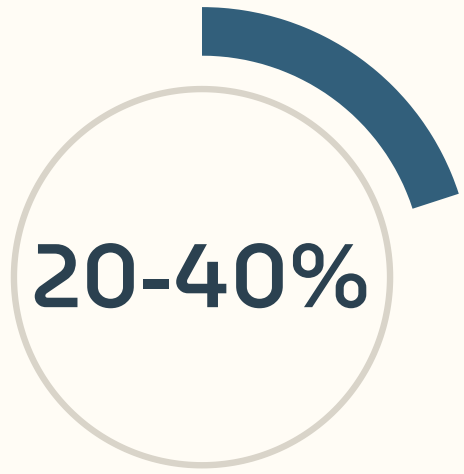
Engage 'Sometimes'

of U.S. hospitals participate in all four interoperability domains — but often only sometimes, not routinely. — ONC/ASTP, 2024



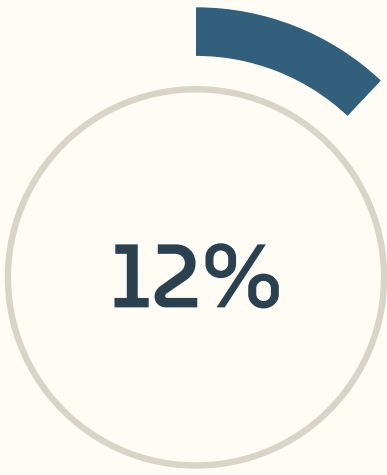
Cross-Continuum Sharing

of hospitals send summary of care records to most long-term/post-acute care providers. — HealthIT.gov Data Brief #71, 2024



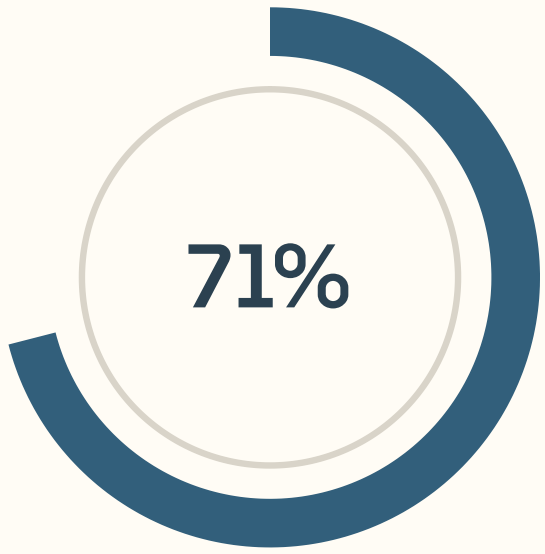
Healthcare Waste

of the $4.3T U.S. annual healthcare spend is classified as waste, much driven by data fragmentation. — Industry estimates, 2025



Duplicate Records

of clinic medical records are duplicates, with 14% containing incorrect patient data. — EHRIntelligence



FHIR Adoption

of global healthcare stakeholders actively use FHIR (up from 66% in 2024), yet version fragmentation creates new challenges. — 2025 State of FHIR Survey

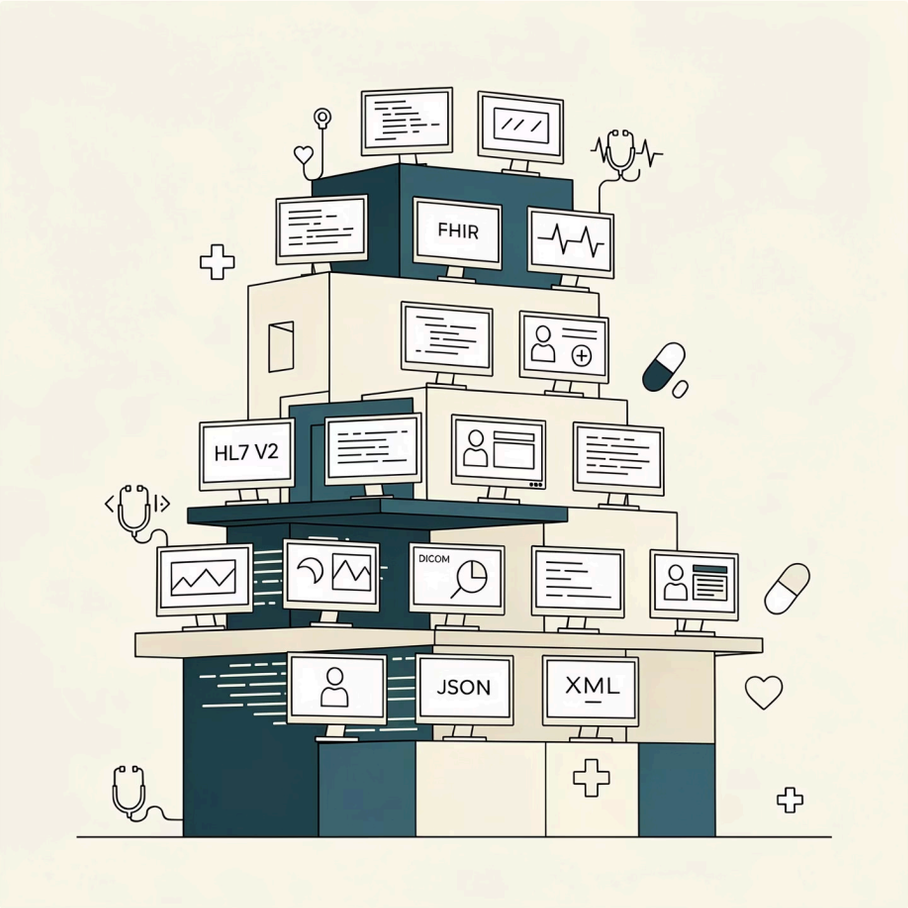
Sources: ONC/ASTP 2024 Data Briefs, HealthIT.gov, 2025 State of FHIR Survey, EHRIntelligence, BlueBriX industry analysis.

PITFALL #1

Lack of Common Standards — The Tower of Babel Problem

Healthcare systems often speak entirely different "languages" due to proprietary data formats and incompatible protocols. EHR vendors frequently represent the same patient data in wildly different structures — blocking seamless exchange at every turn. In 2025, 71% of global healthcare stakeholders actively use FHIR, yet version fragmentation across R4, R4B, R5, and R6 creates new compatibility challenges (2025 State of FHIR Survey). Meanwhile, nearly half (46%) of Health Information Exchange Organizations still need to map non-standard lab codes to LOINC standards (ASTP Data Brief #75, 2024).

Mitigation: Adopt universal standards like **HL7 FHIR** to enable consistent, API-driven data sharing across all systems and vendors.



PITFALL #2

Data Silos — Islands of Information

Departments and organizations hoard data in isolated systems, preventing coordinated care delivery. Many hospitals operate with fragmented data repositories that severely limit clinical decision support and cross-functional collaboration. Only 16% of hospitals send summary of care records to most long-term or post-acute care providers, and just 17% share with behavioral health providers — revealing how deep the silo problem runs across the care continuum (HealthIT.gov Data Brief #71, 2024).

Solution: Enterprise-wide integrated data platforms and cloud infrastructure break down silos and unify access — enabling a single source of truth for patient records.

PITFALL #3

Security and Privacy Concerns — The Double-Edged Sword

Strict regulations like HIPAA create significant friction in sharing sensitive patient data securely. Fear of breaches and costly legal penalties frequently stalls promising interoperability initiatives before they gain traction.

Mitigation: Implement security as an enabler, not a blocker — embed end-to-end encryption, role-based access controls, data anonymization, and automated compliance workflows directly into the interoperability architecture from day one.

PITFALL #4

Lack of Executive Support — The Silent Killer

Leadership often treats interoperability as a technical IT project rather than a strategic organizational priority. Without a C-suite champion, initiatives lack visibility, cross-departmental coordination falters, and projects lose momentum — becoming yet another abandoned initiative.

The Problem

No C-suite champion means no organizational priority, no cross-departmental alignment, and no accountability for outcomes

The Remedy

Build a compelling business case positioning interoperability as a strategic differentiator, regulatory imperative, and competitive advantage — then assign executive ownership

PITFALL #5

Overly Rigid or Static Roadmaps — Failing to Adapt

Cautionary Tale

The EU's INSPIRE Directive failed by locking in standards that couldn't evolve with technology — leaving organizations stuck with outdated frameworks.

A Better Approach

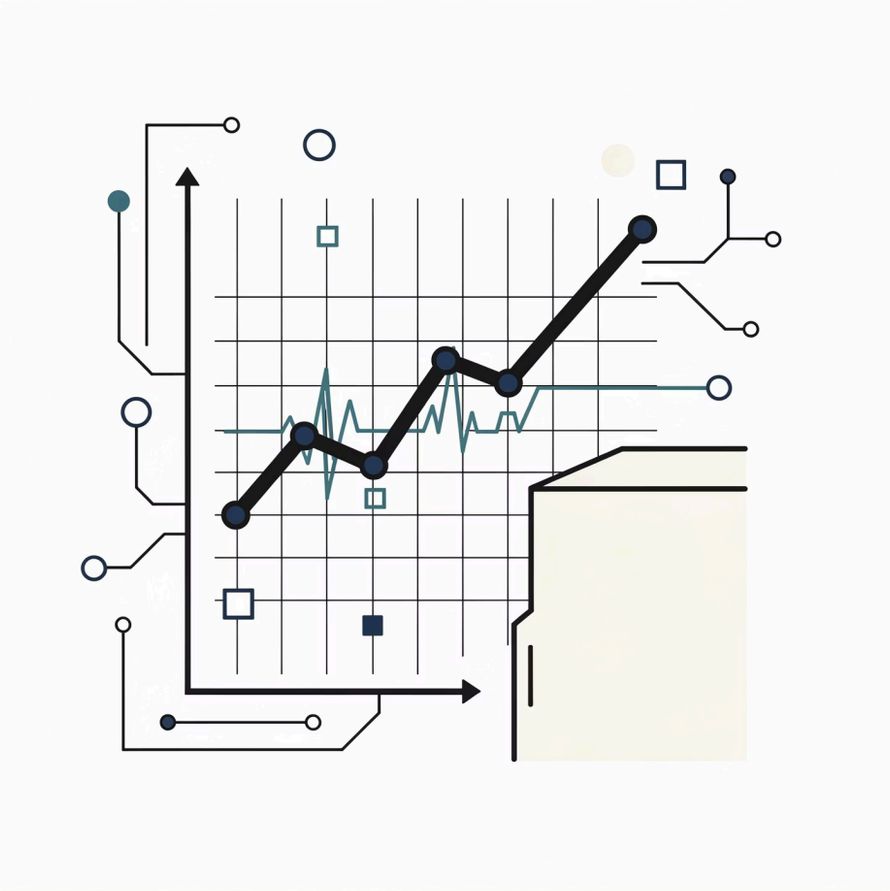
Interoperability demands ongoing negotiation among stakeholders and flexible frameworks. Design roadmaps as **living documents** with iterative updates, regular stakeholder engagement, and built-in mechanisms for adaptation.

$ PITFALL #6

Underfunded Operations — The Budget Cliff Problem

Most interoperability funding targets initial implementation, neglecting crucial ongoing costs. This creates a "budget cliff" where maintenance, training, and essential updates are unfunded after deployment. Without sustained investment, systems stagnate, staff remain untrained, and early interoperability benefits quickly erode, hindering long-term success. With 20–40% of the $4.3 trillion U.S. healthcare spend classified as waste — much of it driven by data fragmentation and redundant processes — underfunding interoperability operations has an outsized economic impact (BlueBriX/industry estimates, 2025).

Mitigation: Build comprehensive Total Cost of Ownership (TCO) models that include ongoing maintenance, training, and change management. Advocate for sustained operational funding, moving beyond one-time implementation budgets.



THE FRAMEWORK

A Framework for Success: Building Roadmaps That Deliver

01

Shared Vision

Align strategy with clinical workflows and patient outcomes from day one

02

Open Standards

Prioritize widely supported standards like FHIR and TEFCA for broad adoption

03

Scalable Infrastructure

Invest in platforms that support real-time data integration and access

04

Security by Design

Embed privacy protections and continuous compliance monitoring from the start

05

Executive Sponsorship

Secure leadership buy-in with measurable ROI and regulatory alignment

06

Sustainable Funding

Plan for total cost of ownership — budget for ongoing maintenance, training, and operational evolution

07

Living Governance

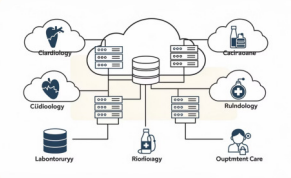
Establish structures for ongoing collaboration and roadmap evolution

SUCCESS STORIES

Real-World Success Stories: From Fragmentation to Flow

TEFCA-Aligned Transformation

Hospitals increased routine interoperability engagement from 30% to over 70% by adopting TEFCA-aligned frameworks and standardized exchange protocols.



Cloud-Based Integration

Integrated cloud data platforms enabled multi-departmental access, dramatically improving care coordination and reducing duplicated diagnostic testing.

Security-First Approach

Organizations that embedded security and compliance early in their roadmaps avoided costly delays, regulatory penalties, and data breaches.

Sustained Funding Model

Health systems that built TCO-based funding models into their roadmaps maintained operational excellence post-implementation, avoiding the common budget cliff that derails long-term interoperability gains.

YOUR NEXT STEP

Take Action: Transform Your Interoperability Roadmap Today

Don't let your roadmap become another failed project. The future of connected, patient-centered care depends on roadmaps that actually work — make yours one of them.

Assess

Evaluate your current state and stakeholder needs

Align

Embrace standards, break silos, and secure leadership

Partner

Engage experts across technical, regulatory, and organizational domains

Evolve

Build flexible, sustainably funded strategies that adapt, train, and deliver long-term value

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